

AUTOMATIC DRIVING COMPUTING

14. PREDICTION (1)

국민대학교
KOOKMIN UNIVERSITY

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지난 시간 복습

OBJECT TRACKING (2)

1. **Kalman filter**

2. **SORT algorithm**

3. **Metric for object tracking**

목차

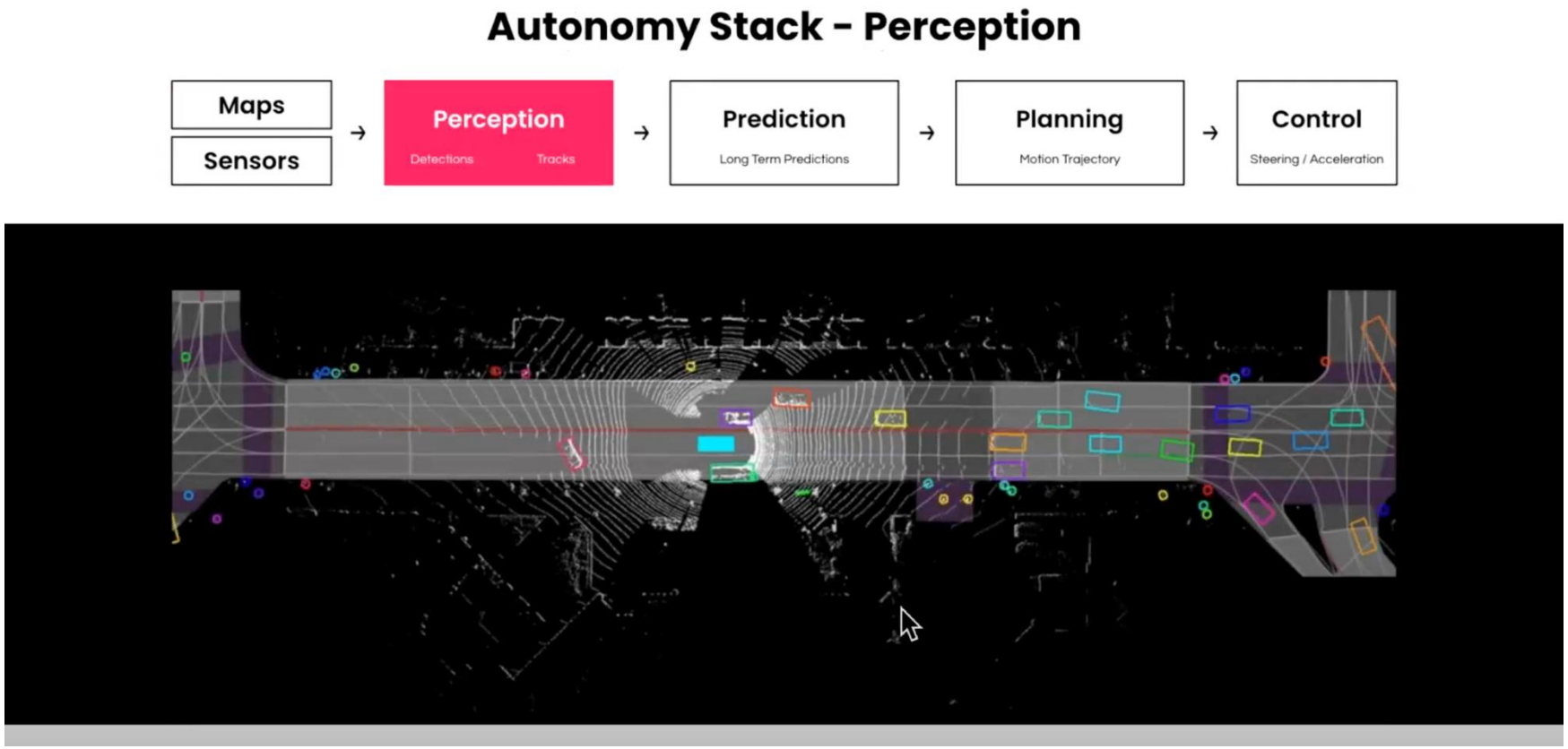
목차

1. **Prediction – overview**

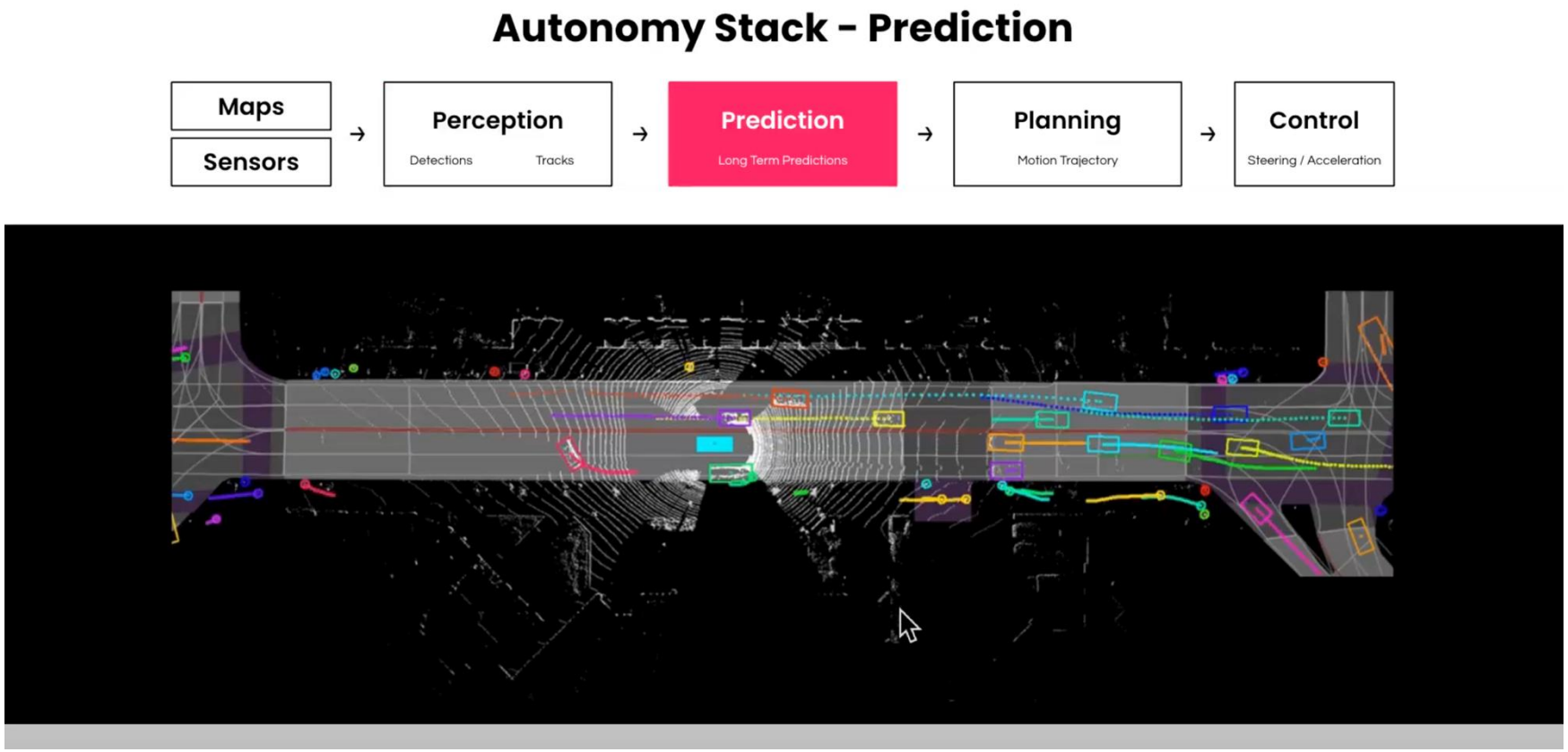
2. **Prediction using raster representation**

PREDICTION – OVERVIEW

6 STACKS OF AUTONOMOUS DRIVING



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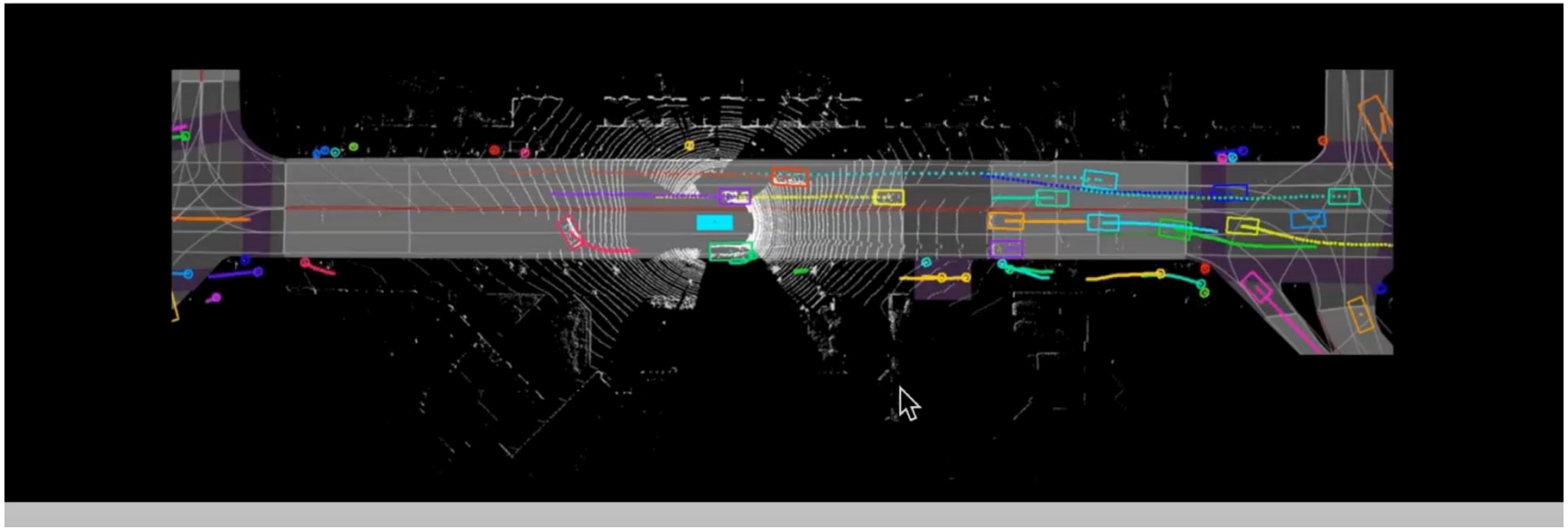


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PREDICTION – OVERVIEW

Goal : 주위 물체들의 미래 이동 경로를 예측하는 것

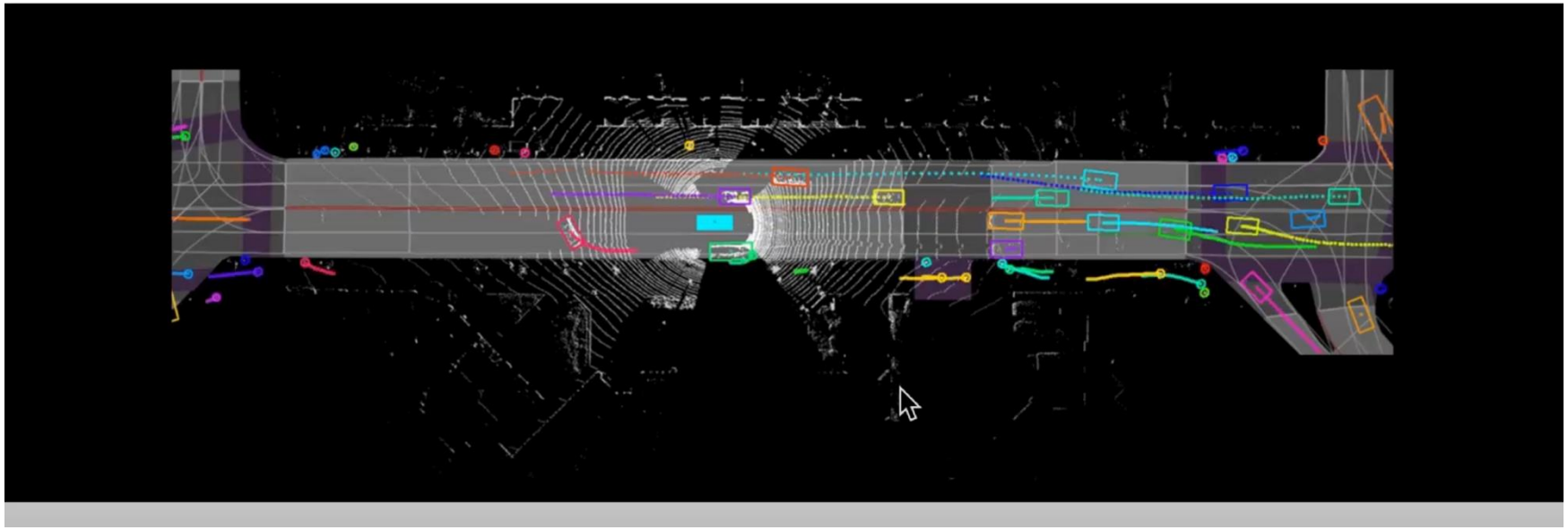
Assumption : Perception (Object detection & tracking) 은 완료됨



PREDICTION – OVERVIEW

Challenge 1 : 주위 물체(객체)들 간에 발생하는 상호작용을 고려해야 함

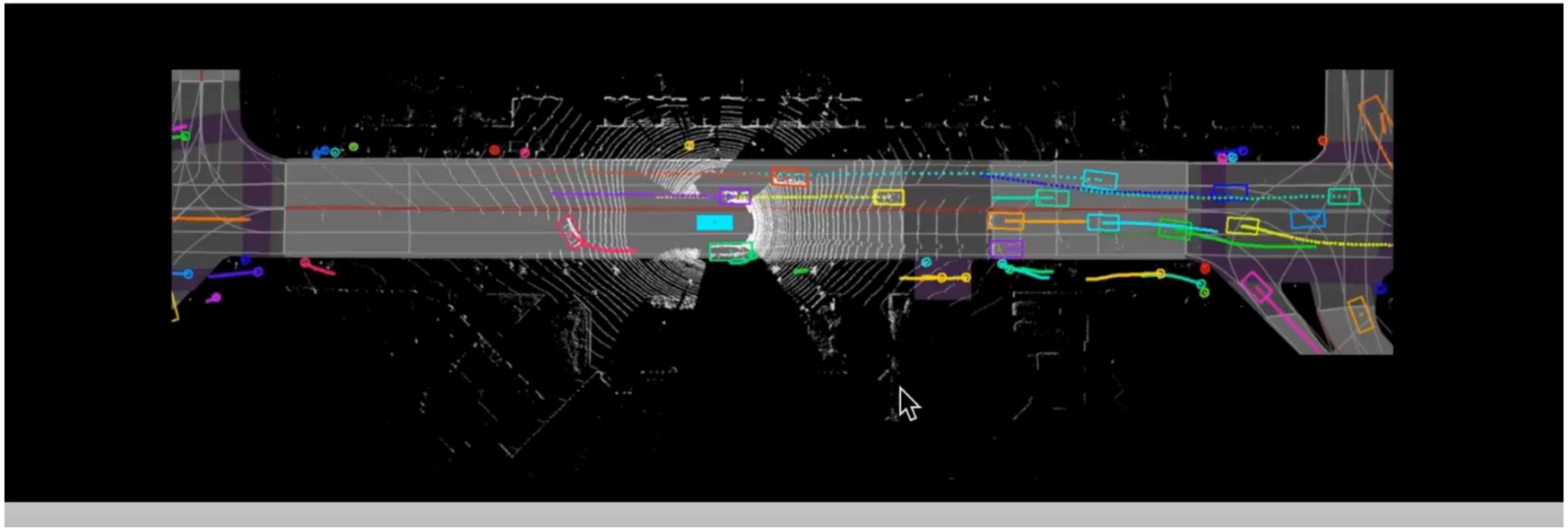
cf) Perception은 물체별로 개별적으로 진행했음



PREDICTION – OVERVIEW

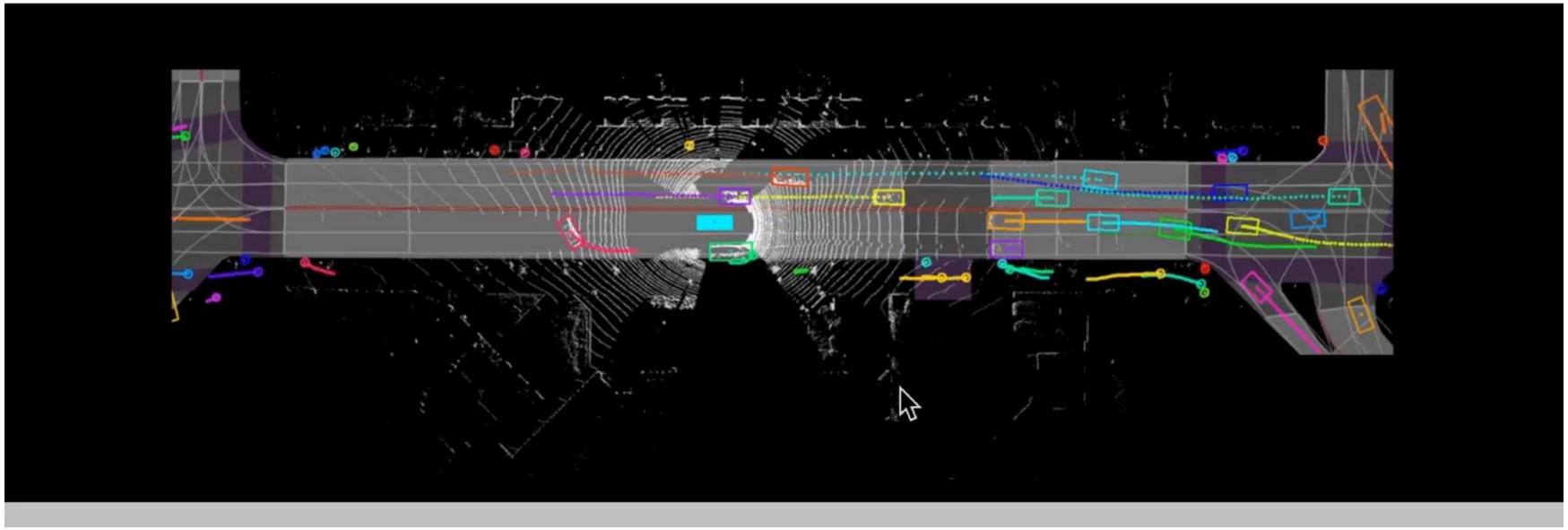
Challenge 2 : 차선, 횡단보도 등의 도로 정보를 고려해야 함

cf) 자율주행 차량은 대부분 SLAM 사용하지 않고 미리 구축된 HD map 사용



PREDICTION – OVERVIEW

Challenge 3 : 미래를 예측 ➔ 100% 정확한 예측 불가능, 불확실성 존재 (Uncertainty)



PREDICTION USING RASTER REPRESENTATION

RASTER

Raster vs Vector



RASTER REPRESENTATION

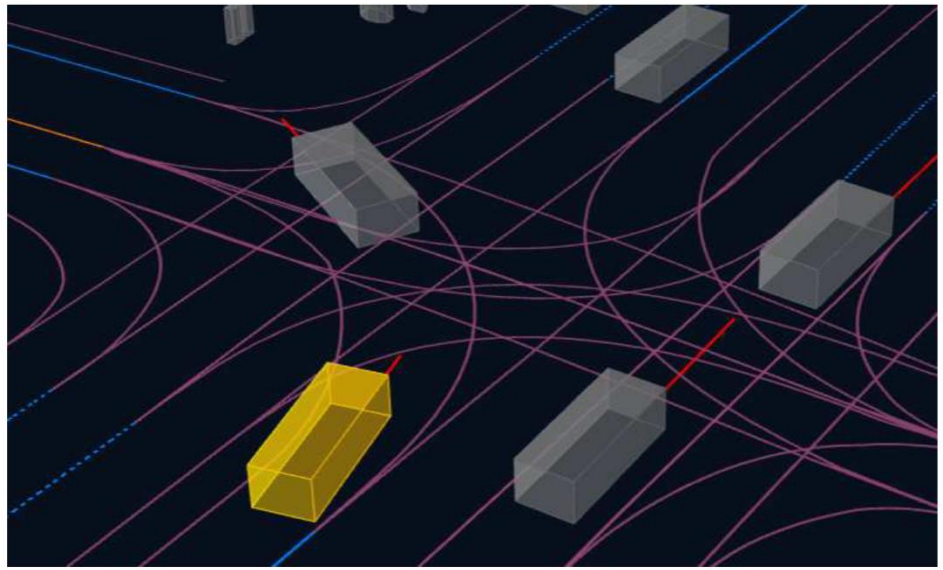
Uncertainty-aware Short-term Motion Prediction of Traffic Actors for Autonomous Driving (WACV 2020)

Uncertainty-aware Short-term Motion Prediction of Traffic Actors for Autonomous Driving

Nemanja Djuric, Vladan Radosavljevic, Henggang Cui, Thi Nguyen,
Fang-Chieh Chou, Tsung-Han Lin, Nitin Singh, Jeff Schneider
Uber Advanced Technologies Group

RASTER REPRESENTATION

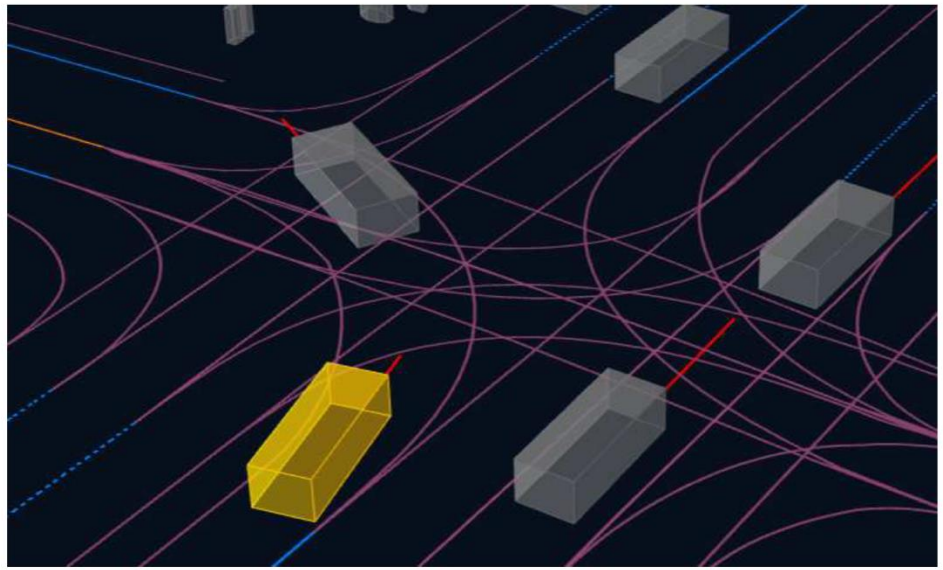
Assumption : High-definition map 및 Real-time tracking system 보유



RASTER REPRESENTATION

Assumption : High-definition map 및 Real-time tracking system 보유

Q) 노란 물체의 미래 이동 경로에 영향을 주는 주변 환경(Context)를 어떻게 표현?

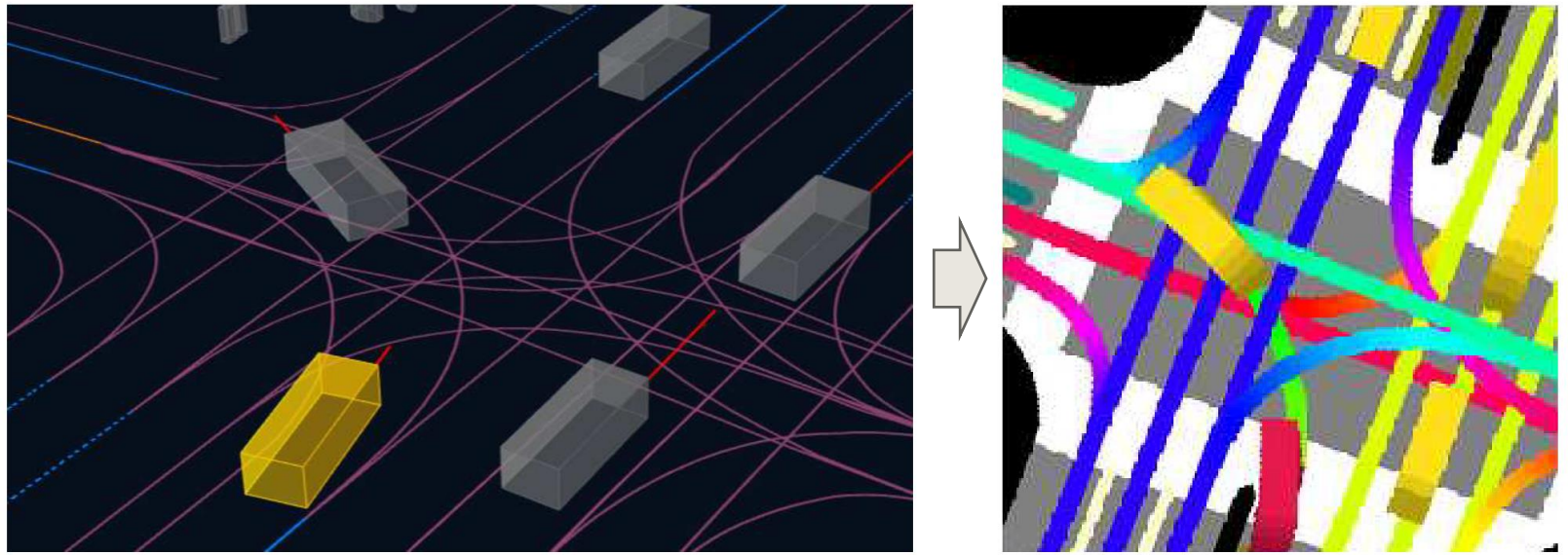


RASTER REPRESENTATION

Assumption : High-definition map 및 Real-time tracking system 보유

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A) (CNN에 입력으로 넣어 줄 수 있는) Raster (2D image)로 표현



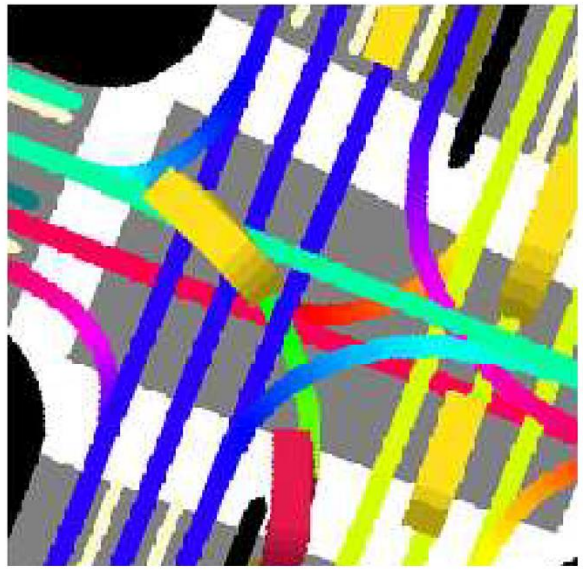
RASTER REPRESENTATION

1) 300 x 300 크기의 rasterized image 생성

pixel resolution : 0.1m

actor of interest : (250, 150) 지점에 위치, 빨간 색 box로 표현

차량의 진행 방향이 위쪽을 향하도록 이미지를 회전

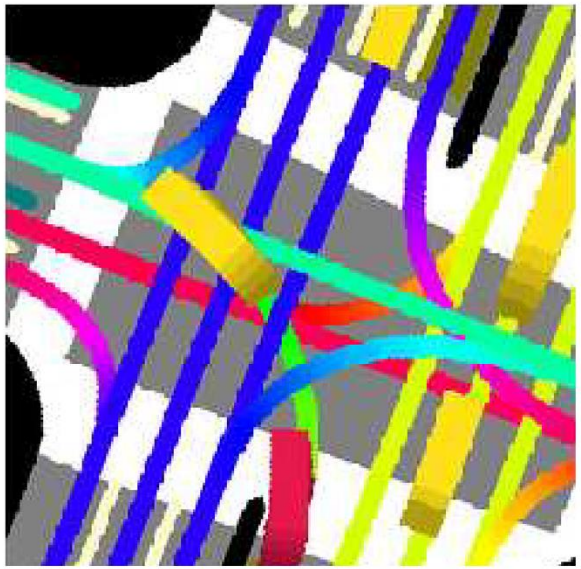


RASTER REPRESENTATION

2) From HD map

횡단보도는 흰색, 차도는 회색, 인도는 검은색으로 표현

차선은 방향별로 다른 색으로 표현

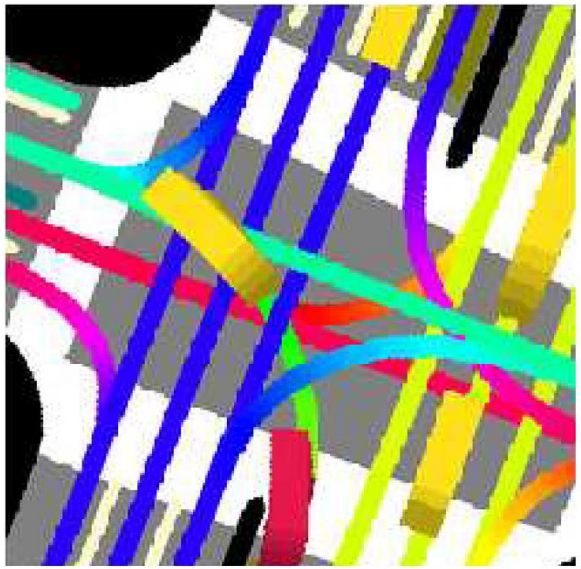


RASTER REPRESENTATION

3) From tracking system

주위 다른 actor들을 노란색 box로 표현

previous time step에서의 다른 actor들 위치는 밝기 정도를 낮춰서 표현 (fading effect)



MODEL TRAINING

Displacement error – Euclidean distance

i : actor index

j : 현재 time step

x, y : ground truth

$\hat{x}, \hat{y}$: model prediction

$$d_{i(j+h)} = \Big(\big(x_{i(j+h)} - \hat{x}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta)\big)^2 + \big(y_{i(j+h)} - \hat{y}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta)\big)^2 \Big)^{1/2}$$

MODEL TRAINING

현재 time step으로부터 1, 2, ..., H time 이후의 경로를 동시에 예측

$$L_{ij} = \frac{1}{H} \sum_{h=1}^{H} d_{i(j+h)}^2$$

$$d_{i(j+h)} = \Big(\big(x_{i(j+h)} - \hat{x}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta) \big)^2 + \big(y_{i(j+h)} - \hat{y}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta) \big)^2 \Big)^{1/2}$$

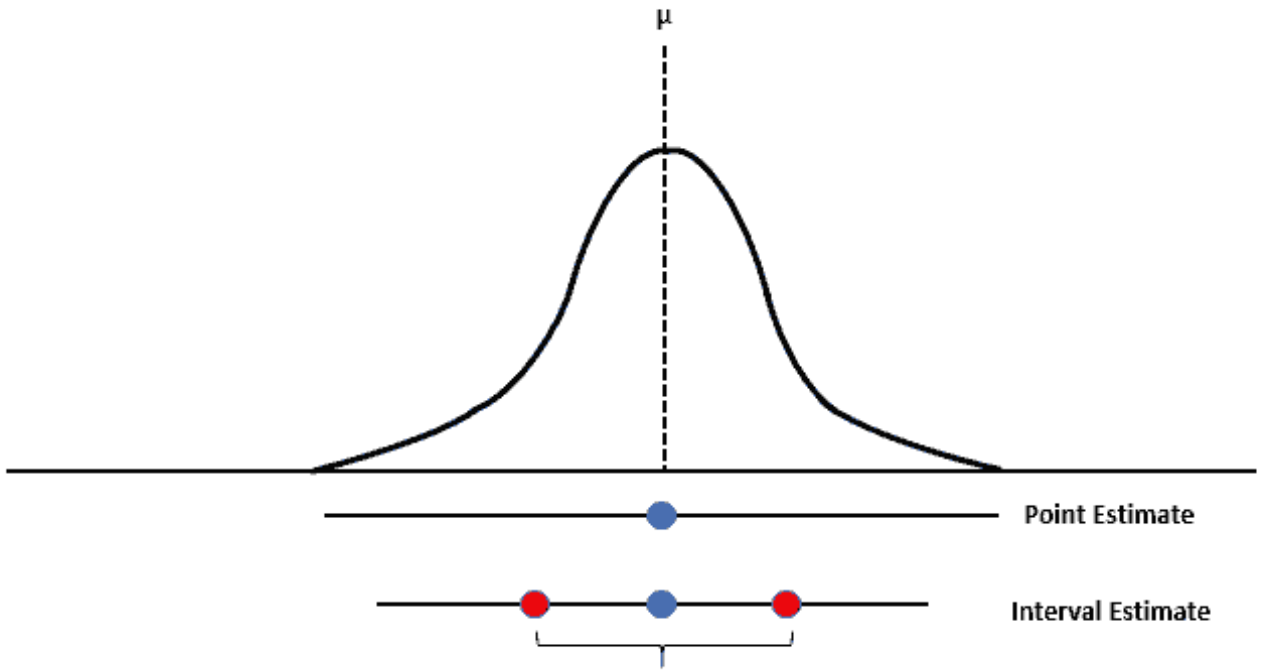
MODEL TRAINING

미래를 예측 ➔ 불확실성 고려 필요

Point estimate vs Interval estimate

Point estimate : 평균 (μ) 만 예측

Interval estimate : 평균 (μ) + 표준편차 혹은 분산 (σ) 을 함께 예측



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MODEL TRAINING

Loss function

Model에서 예측한 sigma 값으로 loss weighting

& Regularization term (sigma 값이 너무 크게 예측되지 않도록)

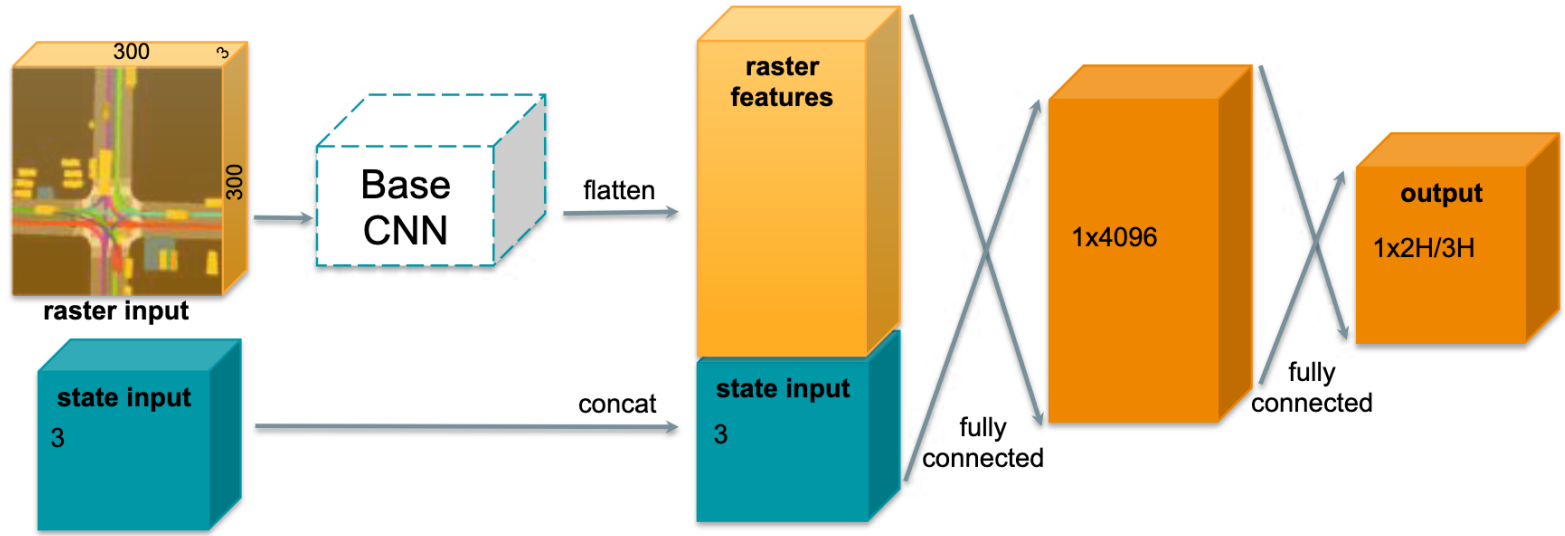
$$L_{ij} = \sum_{h=1}^{H} d_{i(j+h)}^2$$

$$L_{ij} = \sum_{h=1}^{H} \left(\frac{d_{i(j+h)}^2}{2\,\hat{\sigma}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta)^2} + \log \hat{\sigma}_{i(j+h)}(\mathcal{S}_j, \mathcal{M}, \theta) \right)$$

MODEL ARCHITECTURE

Cf) state input : Actor of interest의 속도, 가속도, 방향 변화율

Output : uncertainty 고려할 경우 3H, 안할 경우 2H

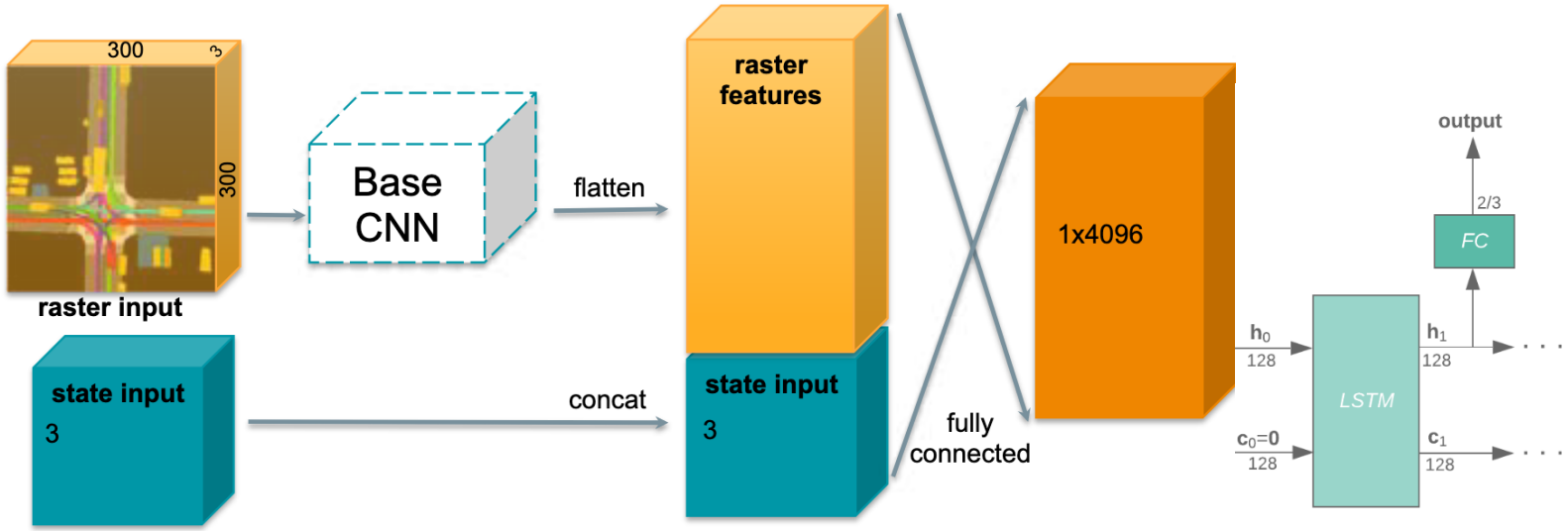


MODEL ARCHITECTURE

Output : uncertainty 고려할 경우 3H, 안할 경우 2H

한번에 FC layer로 예측할 수도 있고,

LSTM으로 매 timestep별로 예측할 수도 있음



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SUMMARY

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1. **Prediction – overview**

2. **Prediction using raster representation**

NEXT LECTURE

Prediction (2) - Prediction using vector representation

QUESTIONS?